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Samsung Innovation Campus, Artificial Intelligence Course
Capstone Idea Proposal

FATURA DEDEKTIFI

Bill Detective

WHICH APPLIANCE USES THE ELECTRICITY AT HOME?

THE BILL SHOWS ONE NUMBER

Electricity bill

TOTAL	one number, in kWh
Fridge	?
Washing machine	?
Dishwasher	?
Kettle	?
Microwave	?

It does not say how much each appliance used.

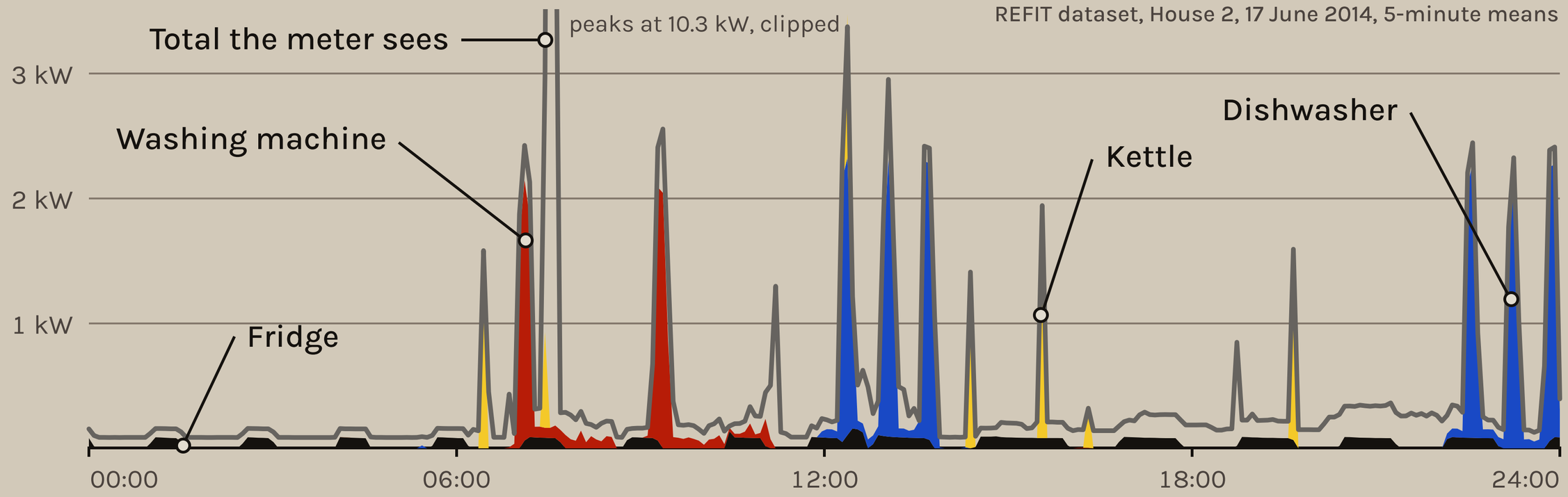
19.5%

of people in Türkiye cannot keep their home warm enough.

Eurostat, 2025

A separate meter on every appliance is expensive and impractical.

ONE DAY IN A REAL HOME

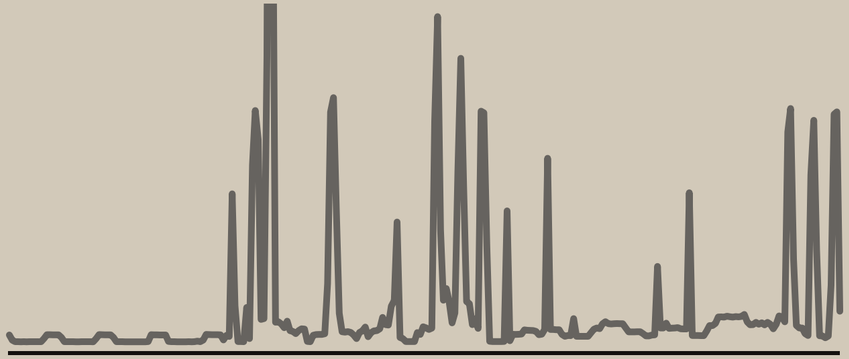


THE DISHWASHER USED A QUARTER OF THAT DAY'S ENERGY

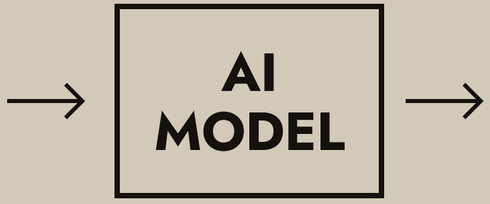
2.72 kWh	1.15	0.98	0.67	4.82
Dishwasher	Washer	Fridge	Kettle	Other appliances

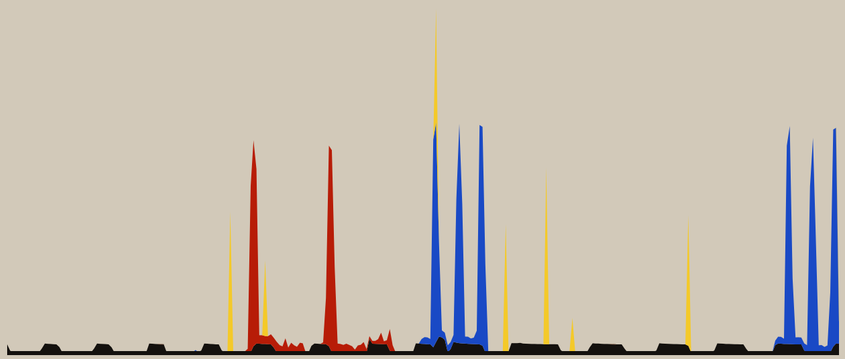
Day total 10.34 kWh

COLOURS FROM THE GREY LINE



Input: the meter’s total signal





Output: use per appliance



Appliances run at the same time.

Their power levels look alike.

2.1 kW
Washing machine heater

2.2 kW
Dishwasher heater

SUSTAINABLE DEVELOPMENT



7.3 ENERGY EFFICIENCY

People can only reduce the consumption they can see.

Indirect contribution

SDG 12	Responsible consumption and production
SDG 13	Climate action

TWO KEY STUDIES

KELLY AND KNOTTENBELT, 2015

Deep neural networks beat classical methods.

“Neural NILM”, ACM BuildSys

ZHANG ET AL., 2018

From a window of the signal, one appliance’s power at that moment.

“Sequence-to-point learning”, AAAI

Missing from both abstracts

A TOOL FOR HOUSEHOLDS

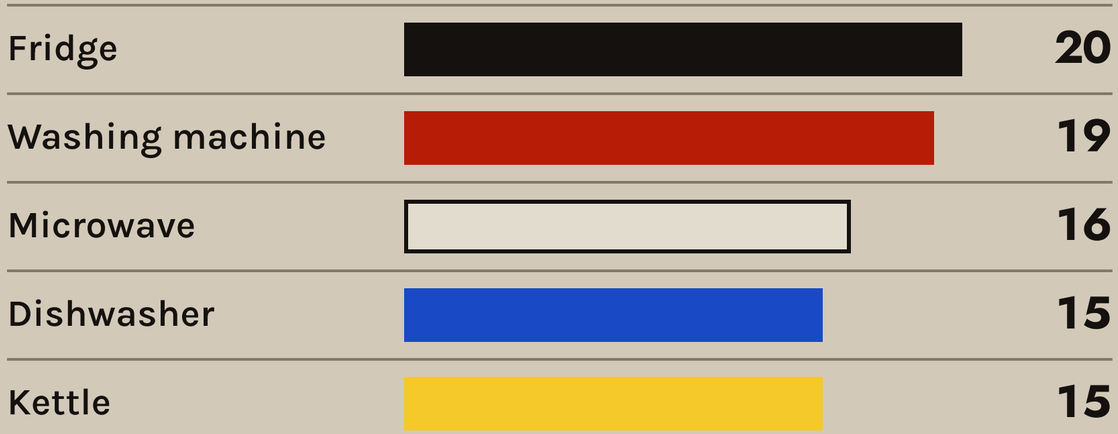
A COST CALCULATION

This project: tested on homes it has never seen, results in kWh and lira.

THE REFIT DATASET

20 HOMES	CC BY 4.0	8 SEC	119.5 MILLION
UK, 2013-2015	Open licence, free	About this often: total power, 9 appliances	Rows, about 6.9 GB

Measured in how many homes



25%

A quarter of one home’s minutes have no data.



Plan: clean the gaps, split the homes into training and test sets.

METHOD AND FIRST TEST

DEEP LEARNING

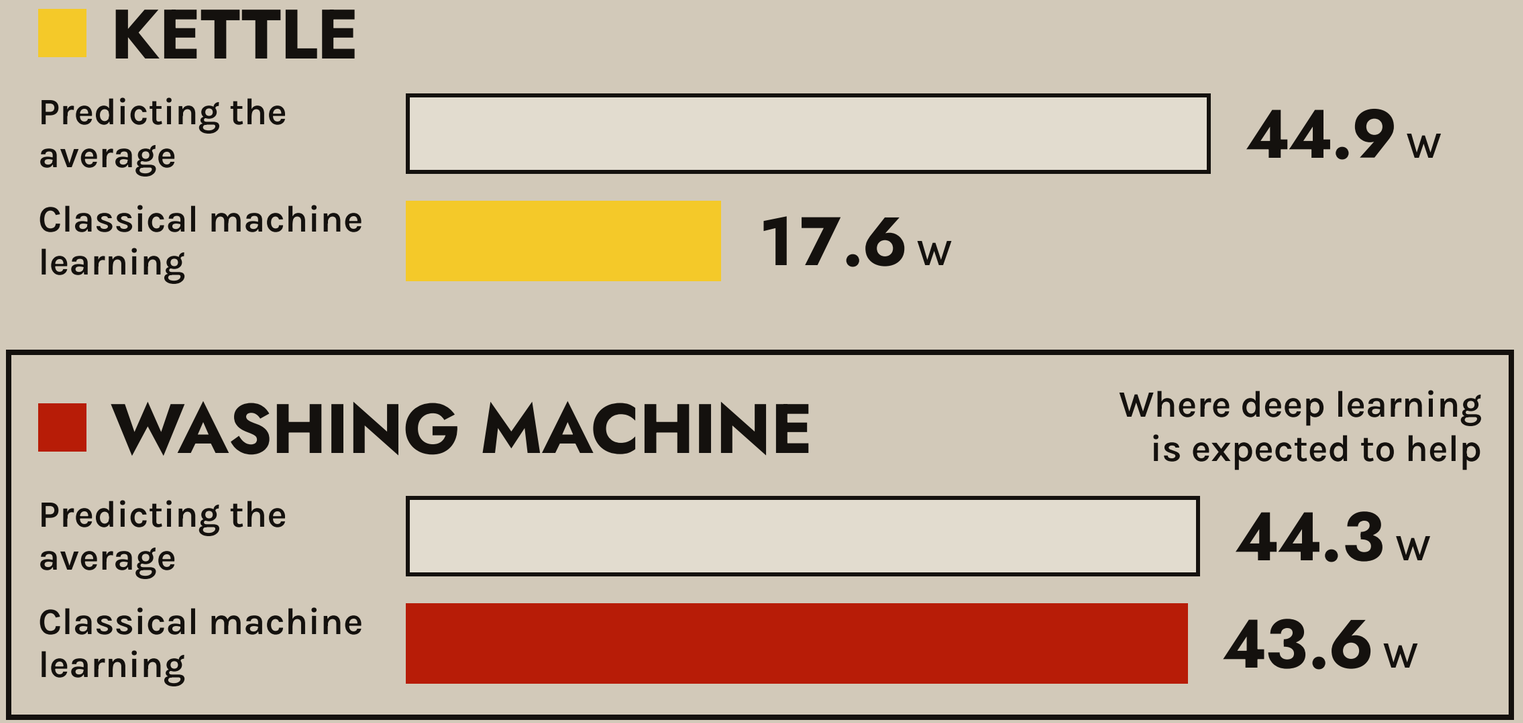
Input: a raw time series.

A small convolutional neural network for each appliance (sequence-to-point).

Target: lower error than the classical method for at least four of the five appliances.

Mean absolute error (watts)

Trained on House 2, tested on House 5



THE BILL IN TÜRKİYE: DAY AND NIGHT TARIFF

Three-rate tariff, TL per kWh

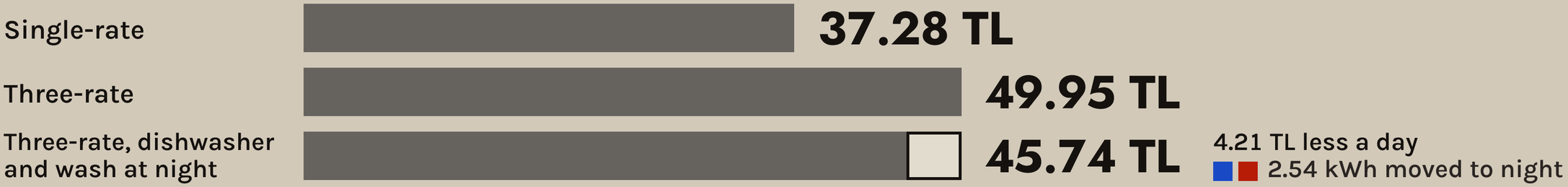


Single-rate, TL per kWh



Prices: EPDK residential tariff valid from 4 April 2026, taxes included by calculation.

The day from slide 3, 10.34 kWh, on both tariffs



For this home the single-rate tariff is cheaper. The demo makes this comparison for every home.

FROM UK DATA TO TÜRKİYE

230 V **50** Hz

The same grid in the UK and in Türkiye

5

The five target appliances are in Turkish homes too

1 TEST ON A SECOND DATASET

Error of a model trained on REFIT

D'Incecco, Squartini and Zhong, 2019

UK-DALE, UK	11.8 W
REDD, US	32.1 W
Trained on REDD, REDD	23.7 W

2 LOADS OF TURKISH HOMES

Missing in REFIT, added to the training data artificially

AIR CONDITIONER **WATER HEATER** **TEA MAKER**

3 MEASURE ON HOURLY DATA

Smart meters in Türkiye give hourly data

1 MIN **15 MIN** **1 HOUR**

4 MEASURE ONE TURKISH HOME

Smart plugs for a few weeks

The error, reported on real Turkish data

DEMO AND LIMITS

AT THE END OF THE TERM A DEMO IN THE BROWSER

Uploaded	Shown
One day of meter data	kWh and lira per appliance, the two tariffs compared, what to move to the night

Limit

I found no open labelled data from Türkiye. My own measurement will be a small check.

SEEING THE DISHWASHER'S SHARE, A FAMILY CAN RUN IT LESS OFTEN

THANK YOU